

COT5520: COMPUTATIONAL GEOMETRY

Homework # 2

Your solutions should be concise, but complete, and typed (or handwritten clearly). Feel free to consult textbooks, journal and conference papers and also each other, but write the solutions yourself and cite your sources. Answer **only five** of the following seven questions. Each problem is worth 20 pts.

1. Implement the following simple polygon triangulation algorithms using a programming language of your choice. Test your algorithms on self-generated random large data sets. Summarize your observations and experiments (ease of implementation, comparison of computation times, etc.). You may use available libraries.
 - (a) $O(n^3)$ -time naive algorithm that relies on the ear-cutting theorem discussed in class.
 - (b) $O(n^2)$ -time improved version of the naive algorithm by pre-computing (and maintaining) separate lists for convex and concave vertices.
 - (c) $O(n \log n)$ -time plane sweep algorithm discussed in class.
2. Let S be a set of n disjoint triangular islands in a planar ocean. Describe a plane sweep algorithm that runs in $O(n \log n)$ time to find a set of $n - 1$ straight bridges with the following properties
 - Each bridge connects an island to another island.
 - The interiors of the bridges do not intersect each other or the islands.
 - Together they connect all islands to each other.
3. Let C be a set of n circles in the plane. Design and analyze a plane sweep algorithm to compute all k intersection points between the circles in $O((n + k) \log n)$ time.
4. In class, we proved by induction that every simple polygon can be triangulated using exactly $n - 2$ triangles. Extend the proof for polygons with holes and give a formula for the number of triangles in the triangulation of a polygon that has a total of n vertices (on its outer and inner boundaries) and k holes. Prove the correctness of your formula.
5. Design and analyze an algorithm that computes in $O(n \log n)$ time a diagonal that splits a simple polygon with n vertices into two simple polygons each with at most $\lfloor 2n/3 \rfloor + 2$ vertices. [*Hint: Explore the properties of the dual graph of a simple polygon triangulation.*]
6. Given a simple polygon P with n vertices and a point p inside it, design and analyze an efficient algorithm to compute the region inside P that is visible from p .
7. The *pockets* of a simple polygon are the area outside the polygon, but inside its convex hull. Let $\mathcal{P}_1$ be a simple polygon with m vertices, and assume that a triangulation of $\mathcal{P}_1$ as well as its pockets is given. Let $\mathcal{P}_2$ be a convex polygon with n vertices. Describe an algorithm to compute the intersection $\mathcal{P}_1 \cap \mathcal{P}_2$ in $O(m + n)$ time.